RING CONTAINER TECHNOLOGIES: TACKLING ENERGY AND WASTE REDUCTION PROJECTS FOR SIGNIFICANT SAVINGS

SECTOR	LOCATION	FINANCIAL OVERVIEW
Industrial	Decatur	$370,000

BACKGROUND

RING Container Technologies operates 17 plastic bottle and container plants across th United States. Their facilities use High-Density Polyethylene (HDPE) and Polyethylene Terephthalate (PET) plastic resin to produce bottles and containers. Most of RING's production-related energy consumption is from electricity used to operate chillers, cooling towers, and compressed air systems. The company is focused on reducing its electricity usage by incorporating established best practices that address chiller energy use and compressed air leaks. RING implemented an energy and waste efficiency project at its Decatur, Illinois facility and is scaling this approach to additional sites. The project scope included a chiller upgrade, compressed air leak reductions, and waste diversion through plastic waste recycling.

SOLUTIONS

The project started with a chiller replacement at the Decatur, IL, facility and expanded to incorporate multiple energy efficiency and waste reduction measures. Process cooling is very important during the manufacturing of PET and HDPE products to ensure optimal product quality and production rates. The Decatur plant relied on chilled water delivered from inefficient chillers and cooling towers that were past their useful life. Upgrading the chilled water system with more energy-efficient models offered significant energy savings and qualified RING to receive a $26,000 incentive from their utility provider. RING also partnered with a national compressed air leak detection company to audit multiple facilities. Plastic blow molding requires high pressure, sometimes as high as 600 pounds per square gauge (PSIG). Compressed air is therefore the most significant energy user at RING facilities and air leak maintenance yields significant energy savings. The third area RING focused on was plastic waste reduction. As bottles are molded, there is always some leftover resin in the form of "pinch-off" or other material waste. RING sought to divert 100% of this waste from landfills. RING joined Operation Clean Sweep to ensure all production floor waste was being recycled rather than lost to landfills. Operation Clean Sweep, a stewardship program of the Plastics Industry Association (PIA) and the American Chemistry Council's Plastics Division, is designed to keep resin pellet, flake, and powder out of the environment.

The existing chillers at the Decatur, IL, plant operated at a system part load value (SPLV of 0.915 kW/ton. The two new chillers operate at an SPLV of 0.743 kW/ton and have a pressure drop that is 10 PSI lower, reducing pump energy consumption. This resulted ir electrical savings of over 483,000 kWh annually and energy cost savings of $42,000.

Figure 1. Decatur, IL Facility Chillers

Table 1. Decatur, IL Energy Efficiency Chiller Upgrade

Electric Project Calculations and Information			
		Existing or Baseline Equipment	**New Energy Efficient Equipment**
A	**Description of Measure**	287-ton chiller 100hp pump running at 95hp	275-ton HE chiller 100hp pump running at 82hp
B	**Electrical Load (total)**	290.44 kW	239.47 kW
C	**Estimated Annual Hours of Operation**	7,408 hours/year	7,408 hours/year
D	**Estimated kWh/year used**	2,151,590 kWh	1,773,974 kWh

Compressed Air Leak Program:

RING contracted an external company to conduct a company-wide air leak maintenance program. This initiative was supported by internal staff and their local DOE Industrial Assessment Center program , at The University of Memphis, which also assisted with energy assessments and compressed air system evaluations. RING discovered and repaired air leaks estimated at over 1,900 cubic feet per minute total, yielding annual energy savings of over 1.5 million kWh per year and cost savings of $138,000 per year. This program will be repeated annually in addition to day-to-day leak checks conducted by facility staff.

Operation Clean Sweep Initiative:

RING focused on waste diversion as part of its Operation Clean Sweep campaign. Participating plants adhered to a policy of a single stream of resins entering the plants and only two streams leaving – finished product and recycling material resin. All plants have a monthly audit where both the exterior and the interior of the plants are checkec by the Quality Manager to find any resin/production waste that is not being recycled. Plant management is working on providing recycling receptacles close to workstations and improving signage. These results are discussed on a monthly call where best practices are shared, and support is provided to bridge any gaps to ensure that the resin waste is recycled properly.

OTHER BENEFITS

All three projects are highly replicable across RING's portfolio of 17 North American facilities. Air leak maintenance and waste diversion are already rolled out company-wide, and RING has observed a more proactive employee culture around finding and fixing air leaks and properly recycling resin. Additionally, RING is now budgeting to purchase more energy-efficient features for all future chiller and air compressor installations and other major equipment upgrades.

IMAGE GALLERY

DECATUR, IL FACILITY CHILLERS